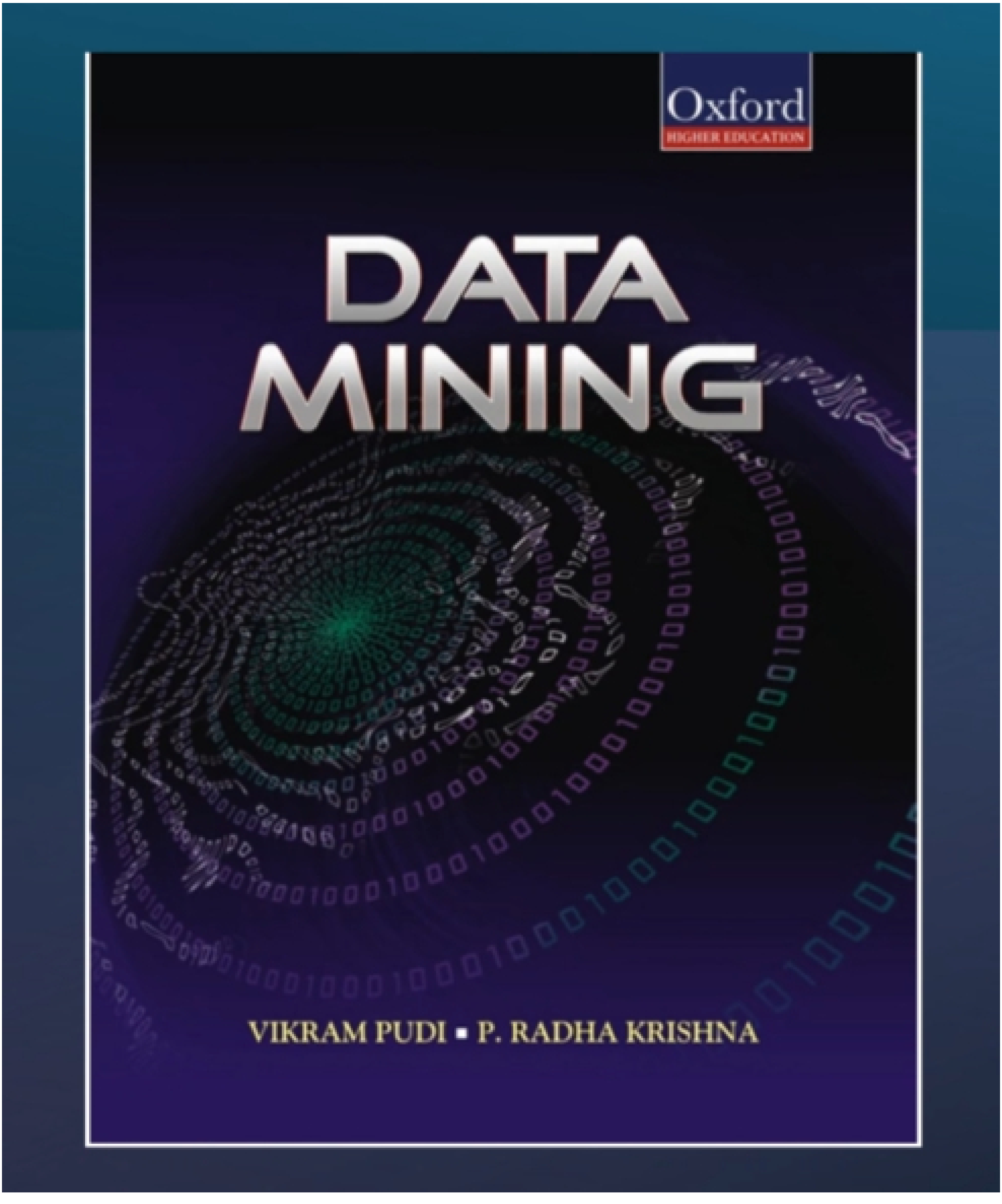
```
Apriori(T, ε)
    L_1 ← {large singleton itemsets}
    k ← 2
    while L_{k-1} is not empty
        C_k ← Generate_candidates(L_{k-1}, k)
        for transactions t in T
            D_t ← {c in C_k : c ⊆ t}
            for candidates c in D_t
                count[c] ← count[c] + 1

        L_k ← {c in C_k : count[c] ≥ ε}
        k ← k + 1

    return Union(L_k) over all k

Generate_candidates(L, k)
    result ← empty_set()
    for all p ∈ L, q ∈ L where p and q differ in exactly one element
        c ← p ∪ q
        if u ∈ L for all u ⊆ c where |u| = k-1
            result.add(c)
    return result
```

Imag

- Different types of association rules (Categorical, hierarchical, cyclic)

- Eclat Algorithm
  - Equivalence Class Transformation: a depth-first search strategy

- FP-Growth
  - Frequent Pattern Growth: a compact data structure called the FP-tree (Frequent Pattern tree) to compress the dataset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Classification

Applications

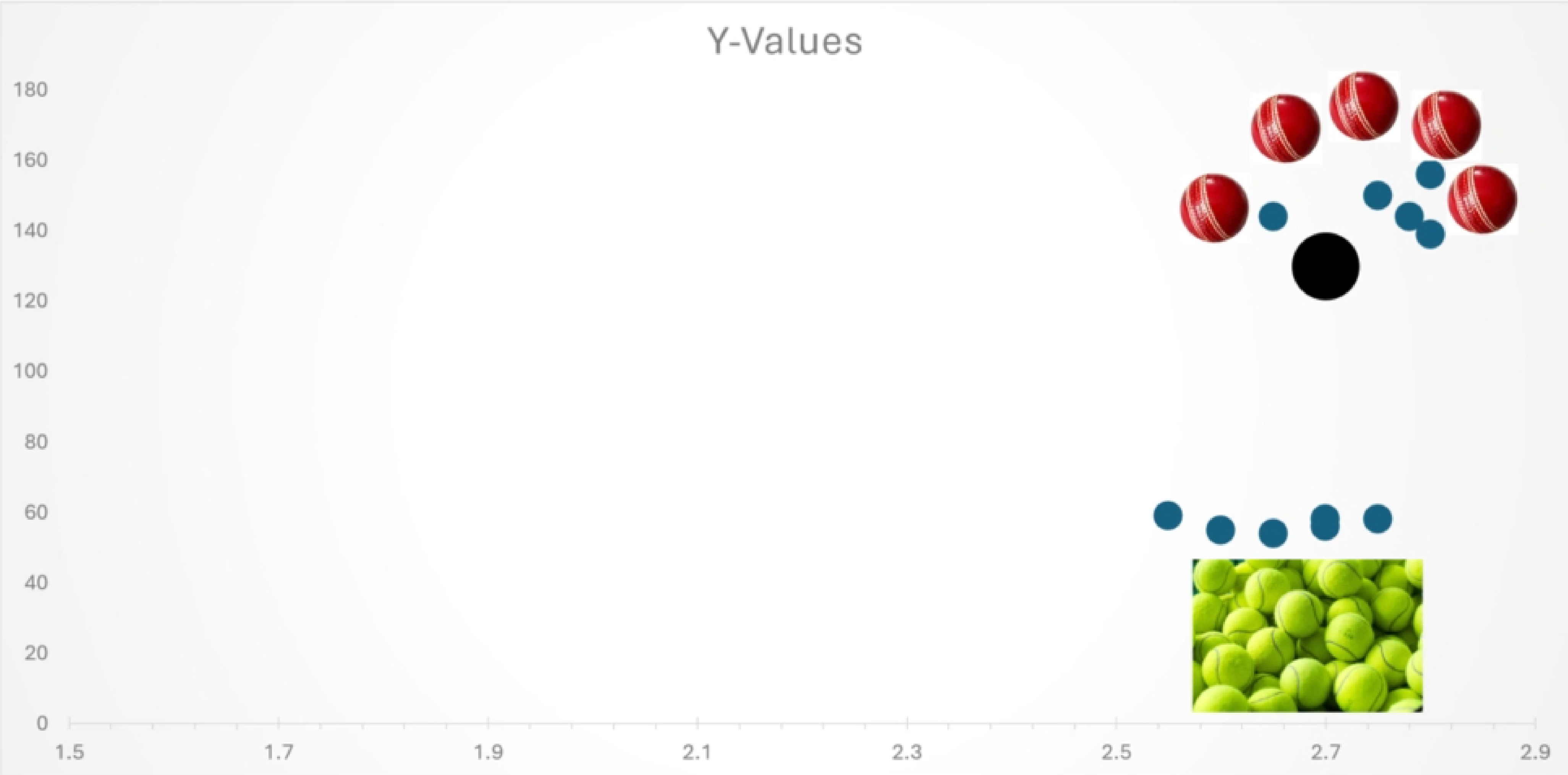
- Text classification
  - Classify emails into spam / non-spam
  - NLP Problems
    - Tagging: Classify words into verbs, nouns, etc.
- Risk management, Fraud detection, Computer intrusion detection
  - Given the properties of a transaction (items purchased, amount, location, customer profile, etc.)
  - Determine if it is a fraud
- Machine learning / pattern recognition applications
  - Vision
  - Speech recognition etc.
- All of science & knowledge is about predicting future in terms of past
  - So classification is a very fundamental problem with ultra-wide scope of applications

- We collect different measurements/facts/about certain features

- $x \ = (x_1, x_2, \dots, x_d)$
  - In the above example, x_1, x_2 are diameter and weight

- $y \in \{1,2,\dots,K\} = [K]$

Cricket ball vs tennis ball



- What is classifier?
- How to measure ‘goodness’ of a classifier?
- If only 1% population has cancer, then a test for cancer that classifies all people as non-cancer will predict 99% of the trails correctly

Bayes Classifier

- Suppose $Y \in \{1,2, \dots, K\}$,
- $P(Y = k)$ Prob that one observes a data point from class k (Prior)

- $P(Y = k|X = x)$ Conditional Probability (the label of the data point point x) $= k$ (Posterior)

- Which class we assign?

- $k \in argmax_{j \in [k]} \Pr(Y = j | X = x)$

- One can argue it is optimal classifier

- What is optimal?

Naïve Bayes Classifier

- What can we say using Bayes theorem for Posterior and Prior?

- $Posterior = \frac{Prior * Liklihood}{P(X=x)}$